Last time

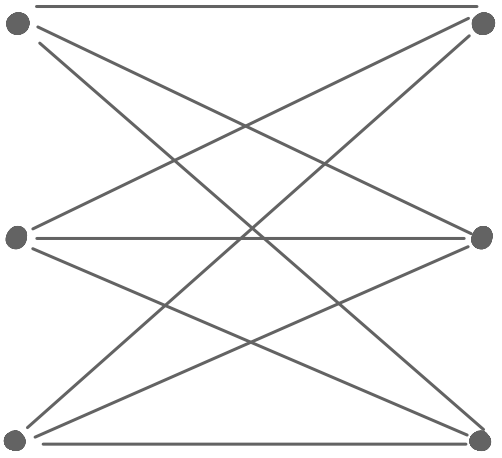
- A planar graph is a graph that be drawn in the plane so that edges don't cross each other.
- A plane graph is a valid drawing of a planar graph.
- Any subgraph of a planar graph is also planar.
- A plane graph G breaks the plane into a collection of connected regions, called faces of G.
- The length of a face is $\text{len}(f) = \#\{\text{edges incident to } f\} + \#\{\text{bridges incident to } f\}$.
- Euler's: If G is a connected plane graph, then $|V(G)| + |F(G)| - |E(G)| = 2$.
- Headshaking: $\sum_{f \in F(G)} \text{len}(f) = 2|E(G)|$.
- If G is planar connected on $n \geqslant 3$ vertices, then $|E(G)| \leqslant 3n - 6$. Hence K_5 is non-planar.
- Moreover, if G is triangle-free then $|E(G)| \leqslant 2n - 4$. Hence $K_{3,3}$ is non-planar.

Definition. *Let* G *be a graph. A subdivision of* G *is a graph obtained by replacing some edges of* G *by paths.*

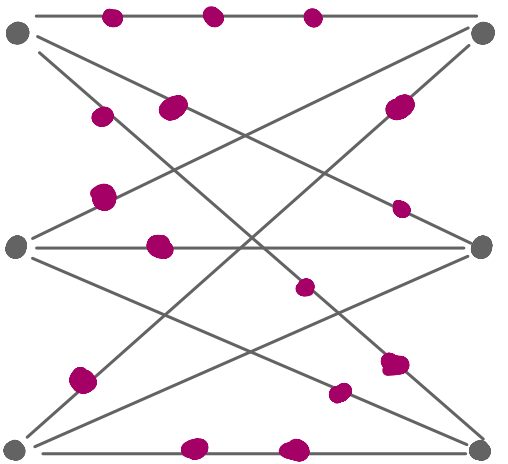
- Informally, a subdivision is obtained by inserting some new vertices into some edges.
- Each vertex introduced when creating the subdivision has degree 2.

Internally disjoint

Example.



$K_{3,3}$



Subdivision of $K_{3,3}$

Lemma. *Let* G *be a graph and* H *a subdivision of* G*. Then* G *is planar if and only if* H *is planar.*

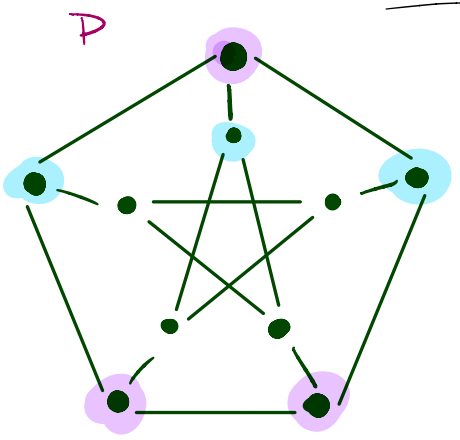
Proof. If G is planar, Then we can fix some valid drawing for G
Then we get a valid drawing for H, just add vertices
in The drawn edges.

If H is planar, and we have a valid drawing for H,
Then To draw The edges of G we can just follow
The paths given by H. ☐



Corollary. *If a graph G contains a subdivision of either K_5 or $K_{3,3}$, then G is non-planar.*

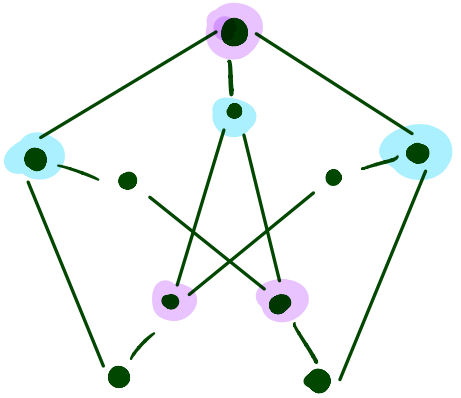
Example (Petersen).



Proof: G planar $\Rightarrow$ every subgraph planar
$\Rightarrow$ subdivision planar
$\Rightarrow$ K_5 or $K_{3,3}$ planar (Lemma)

Can P contain a subdivision of K_5? No, any subdivision of K_5 contains a vertex of degree 4
In P, degrees are 3.

Kuratowski's Theorem, 9.7. A graph G is non-planar if and only if G contains a subdivision of either K_5 or $K_{3,3}$.



Subdivision of $K_{3,3}$ in P

P is non-Planar.

Back to coloring.

$|E(G)| \leq 3n-6$

Theorem (9.3). *If* G *is a planar graph, then* $\delta(G) \leq 5$. $n = |V(G)|$

Proof: If $G_1, \dots, G_k$ are the components of G, $\delta(G) = \min_i \delta(G_i)$

So can assume G connected. Clear if $n=1$, $n=2$. Assume $n \geq 3$

Hence

$$3n-6 \geq |E(G)| = \frac{1}{2} \sum_{v \in V(G)} \deg v \geq \frac{1}{2} \sum_{v \in V(G)} \delta(G) = \frac{1}{2} n \delta(G)$$

(the first $\geq$ is Thm 9.2)

$$\Rightarrow \frac{1}{2} n \delta(G) \leq 3n-6 \Rightarrow \delta(G) \leq \frac{2}{n}(3n-6) = 6 - \frac{12}{n} < 6$$

$$\Rightarrow \delta(G) < 6 \Rightarrow \delta(G) \leq 5.$$

Corollary (6-color theorem). *If* G *is a planar graph, then* $\chi(G) \leq 6$.

Proof: If H is any subgraph of G, then H is planar. Thus $\delta(H) \leq 5$. Then $d(G) = \max\{\delta(H) : H \text{ is a subgraph}\} \leq 5$

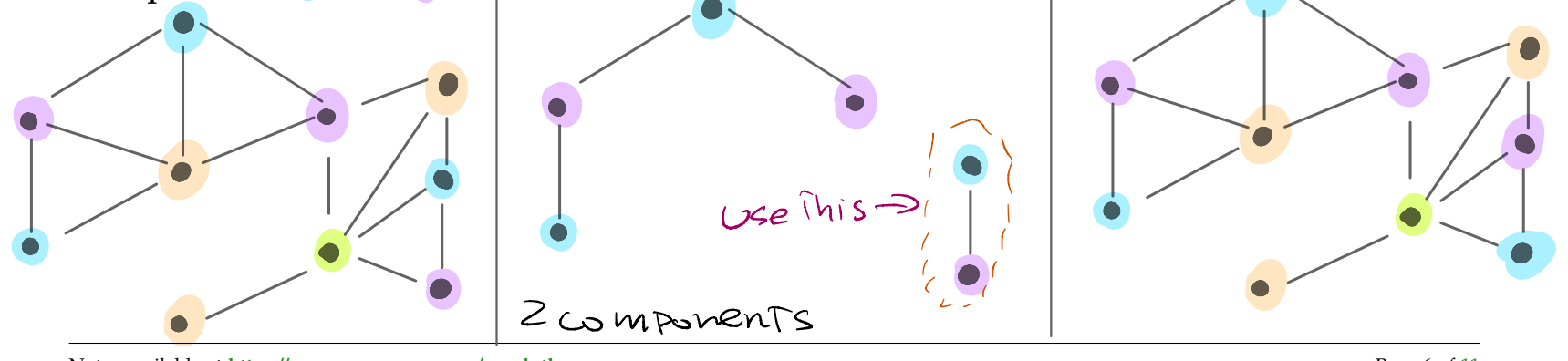
By Thm 10.7, $\chi(G) \leq 1 + d(G) \leq 1 + 5 \leq 6$

Definition Let $f\colon V(G) \to C$ be a proper coloring of a graph G. Fix two colors $a \neq b \in C$.

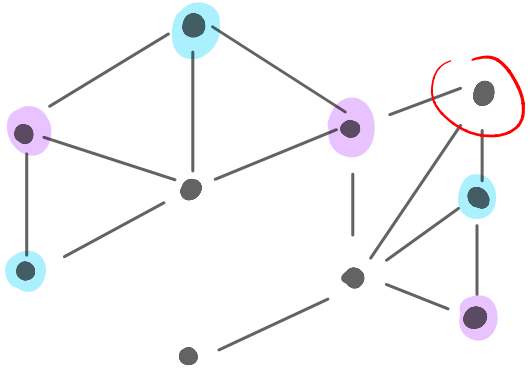
- We use $G[a,b]$ to denote the induced subgraph of G containing all vertices colored by a or b.
- An $[a,b]$-**Kempe component** is a component of $G[a,b]$. We use $C[a,b]$ to denote such components.
- The **Kempe swap** associated to a Kempe component $C[a,b]$ is the coloring $f'\colon V(G) \to C$ where

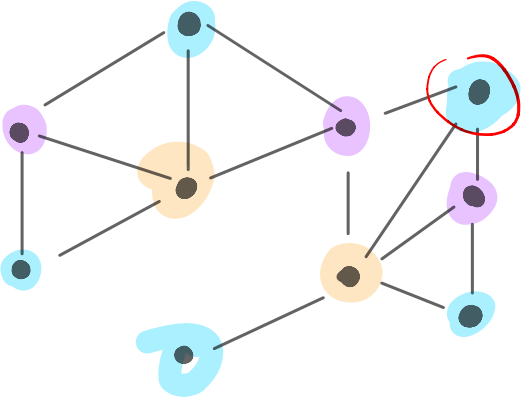
$$f'(v) = \begin{cases} f(v), & \text{if } v \notin C[a,b], \\ a, & \text{if } v \in C[a,b] \text{ and } f(v) = b, \\ b, & \text{if } v \in C[a,b] \text{ and } f(v) = a. \end{cases}$$

Example. $a =$, $b =$



Idea: Suppose we are coloring a graph. Suppose at some point we get to a vertex whose neighbors are using all previous colors. So we need a new color





Instead, perform a swap, which could free up some of the previously used colors.

Lemma. *The Kempe swap f' is a proper coloring.*

Proof:
- f' does not change colors outside $C[a,b]$
- f' is proper in $C[a,b]$ (just a swapping)
- If $xy \in E(G)$, where $x \in C[a,b]$, $y \notin C[a,b]$

Then $f'(x)$ is either a or b.

But $f'(y)$ cannot be a or b, otherwise, since $xy \in E(G)$, y would be in the same component

Thus $f'(x) \neq f'(y)$ ∎

Theorem (5-color theorem). *If* G *is a planar graph, then* $\chi(G) \leqslant 5$.

Proof: Assume G planar but $\chi(G) \geqslant 6$. Want a contradiction.

Step 1. Can assume wlog that G is 6-critical.

By Thm 9.3 since $\chi(G) \geqslant 6$, G contains a 6-critical subgraph

That subgraph is still planar, and $\chi = 6$

Step 2. $\delta(G) = 5$.

Since G is planar, $\delta(G) \leq 5$

By Prop. 4 Lecture 20, $\delta(G) \geqslant 6-1 = 5$ (G 6-critical)

Step 3. Fix a vertex v with $\deg v = 5$. There is a proper coloring $f: V(G - v) \to \{1, \ldots, 5\}$.

Such v exists because $\delta(G) = 5$. Since G-v is a proper subg of a 6-critical graph (G), we know $\chi(G-v) < 6$

Step 4. Can assume wlog that the 5 neighbors of v use distinct colors.

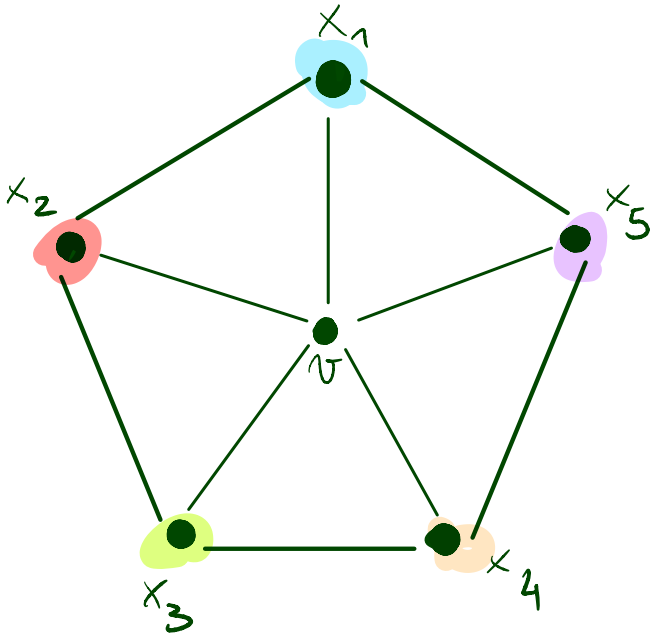
otherwise, if some color in $\{1,\dots 5\}$ is not used in $N(v)$, Then we can use That color for v To get a proper coloring of G with at most 5-colors, contradicts $\chi(G) \geq 6$.

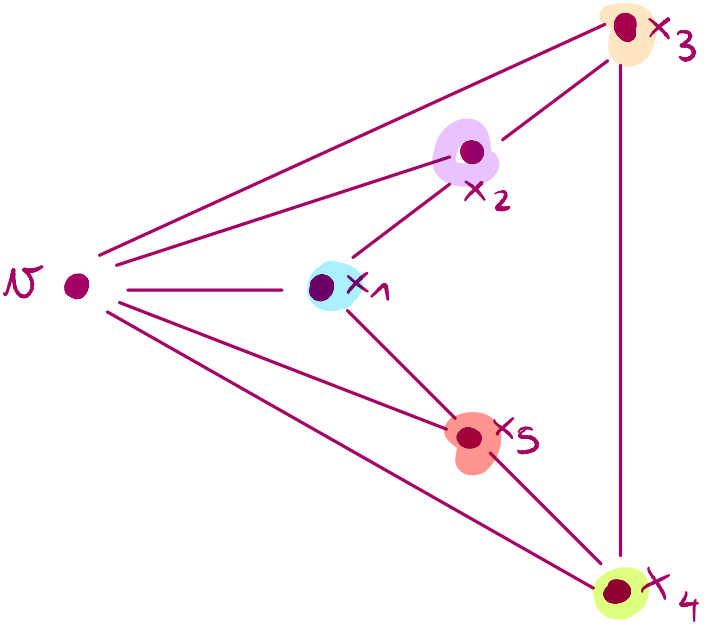
Step 5. Can order the neighbors of v anticlockwise: $x_1, \dots, x_5$ and assume $f(x_i) = i$.

G is planar. Take a valid drawing. There are 5 edges going out of v, order These edges anti clockwise. Then use That ordering for The 5 endpoints. Get $f(x_i)=i$ by relabel of colors.

Step 6. If we add the 5 edges $x_i x_{i+1}$, the new graph is still planar and the coloring remains proper.

Based on The ordering, x_i and x_{i+1} share a common face Then we can draw a line from x_i To x_{i+1} without leaving That face, connecting x_i To x_{i+1}, Thus The new graph will be planar. Still proper coloring by Step 4.


x_1
x_2
x_5
v
x_3
x_4


x_3
x_2
v
x_1
x_5
x_4

$G' = G - v$

Step 7. Let $C[1,3]$ denote the $[1,3]$-Kempe component containing x_1. Then $C[1,3]$ contains x_3.

Otherwise, we perform a Kempe swap to recolor x_1 with color 3 and leave coloring for $x_2, \dots, x_5$ unchanged

Thus we can color v with color 1, to get a proper 5-coloring for G, contradicts $\chi(G) \geq 6$

Step 7.5 There is an x_1-x_3 path $P_{1,3}$ which uses only the colors 1 and 3 and contains no vertices from $v, x_1, \dots, x_5$ other than x_1 and x_3.

By step 7, x_1, x_3 are on the same component of $G'[1,3]$. There is an x_1-x_3 path $P_{1,3}$ in $G'[1,3]$.

Step 8. There is an x_2-x_4 path $P_{2,4}$ which uses only the colors 2 and 4 and contains no vertices from $v, x_1, \dots, x_5$ other than x_2 and x_4.

Step 9. $P_{1,3}$ and $P_{2,4}$ are vertex-disjoint.

Just because they use different colors.

Step 10. G contains a subdivision of K_5.

Consider v, x_1, x_2, x_3, x_4.

We have edges $vx_1, vx_2, vx_3, vx_4, x_1x_2, x_2x_3, x_3x_4$

We are missing x_1-x_3, x_2-x_4, x_4-x_1

We have paths P_{13}, P_{24}, (x_4, x_5, x_1)

Internally disjoint. Gives subdivision of K_5.

Contradiction ☒